SynchroKnot: The New Genre Solution for Decentralised Cloud Computing and Data Centres

Cloud computing and data centres are the backbone of the current global digital economy. There is a standard architecture for implementing cloud computing, which is centralised in terms of location, but disparate in terms of hardware and software. SynchroKnot offers a decentralised software solution for embedded platforms, which allows for complete cloud computing solutions on a virtual System on Chip (vSoC).

Cloud computing allows manipulating, configuring, and accessing hardware and software resources remotely. It makes our business applications mobile and collaborative, but needs an extensive IT infrastructure. A standard infrastructure consists of virtualisation software such as VMware, OpenStack, Hyper-V, RHEV, Xen, VDI, etc. Below the virtualisation layer, these virtual machines run on servers. The data gets stored in SAN/NAS or other distributed block storage. All these systems are connected by redundant switches and routers.

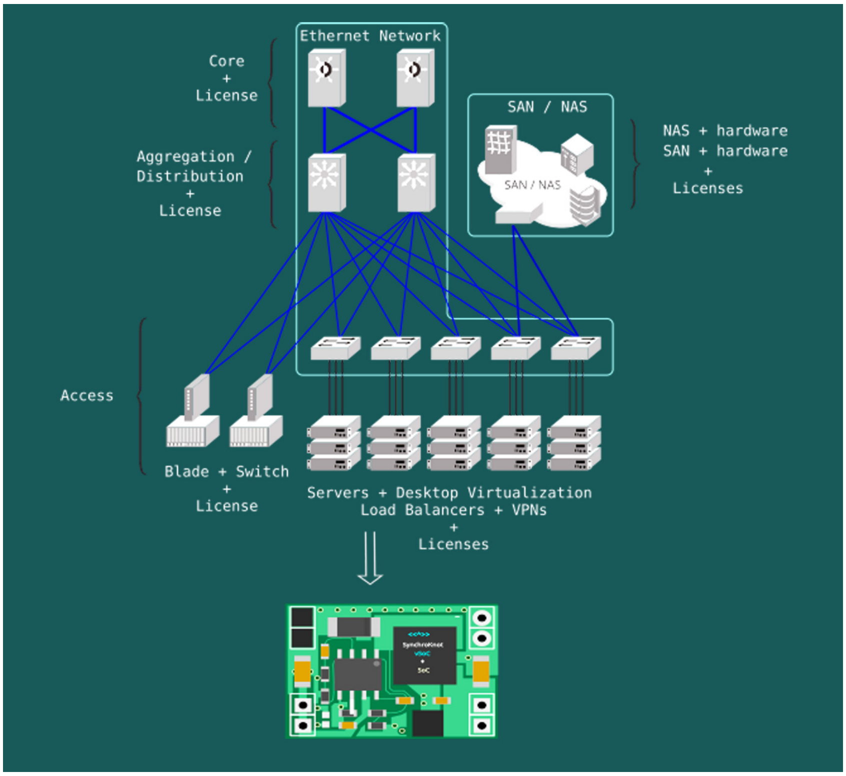

Figure 1: Transforming Centralised Cloud & Data Center architecture into one vSoC + SoC *(Image courtesy of SynchroKnot)*

The challenge with this kind of architecture is that you need to purchase expensive disparate hardware like servers, redundant switches and routers, as well as storage and load balancers. You also need to buy virtualisation software with licences, and licences for hardware and storage. Different hardware and software must be separated through management panels, APIs and user interfaces. Moreover, a large space is needed to house expensive hardware with thermal management solutions, and a whole multi-tiered team is required to manage the complex operations.

The complexity increases further when scaling this architecture. To overcome these difficulties, SynchroKnot, along with ASRock, its hardware partner from Taiwan, has created an elegant alternative to cloud based architecture. The solution consists of a virtual System on Chip (vSoC) which literally acts as a server. Each core of the SoC runs at around 4.2GHz without overclocking. One can run widely accepted tools like OpenStack on this hardware itself. The hardware comes with built-in virtual switches to ensure a standard configuration.

In traditional cloud based architecture, the data gets stored in SAN/NAS. The data in the case of SynchroKnot gets stored in a file